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In [2]: import numpy as np
        from scipy.stats import norm
```

Q1

(a) Compute the implied volatility of the option using the Black-Scholes Model. Use trial-and-error or interpolation. Try at least three volatility levels in the range 0.1 to 0.5. Use the Black-Scholes formula for a European call option.

```
In [3]: def current_value_call(S0, K, r, T, sig):
        # sig is for sigma, the volatility
        d1 = np.log(S0/K) + (r + (sig**2)/2)*T
        d1 /= sig*np.sqrt(T)
        d2 = d1 - sig*np.sqrt(T)

        N1 = norm.cdf(d1, loc=0, scale=1)
        N2 = norm.cdf(d2, loc=0, scale=1)

        C = S0*N1 - K*np.exp(-1*r*T)*N2
        return C
```

```
In [4]: # Calculating IV using quadratic interpolation
C = 4.20 # Current Value of the Option
S0 = 38.0 # Current Stock Price
K = 35 # Strike Price
T = 4.0/12.0 # Time to expiry in years
r = 0.06 # risk-free rate

# Fitting a degree 2 polynomial over these data points and their corresponding current values
proposed_volatilities = np.array([0.1, 0.3, 0.5])
current_values = current_value_call(S0, K, r, T, proposed_volatilities)
coeffs = np.polyfit(current_values, proposed_volatilities, 2)

# IV will be the value of this fitted curve at C = 4.2
IV = np.polyval(coeffs, C)
print(f"Implied Volatility for the given stock is : {IV:.3f}")
```

Implied Volatility for the given stock is : 0.198

(b) Suppose the implied volatility is accepted to be 0.28 from market estimation. Now, using this volatility, calculate the price of a European put option with the same strike and maturity. Use the Black-Scholes formula for a European put option or apply put-call parity.

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In [5]: def current_value_put(S0, K, r, T, sig):
        d1 = np.log(S0/K) + (r + (sig**2)/2)*T
        d1 /= sig*np.sqrt(T)
        d2 = d1 - sig*np.sqrt(T)

        N1 = norm.cdf(-d1, loc=0, scale=1)
        N2 = norm.cdf(-d2, loc=0, scale=1)

        P = -S0*N1 + K*np.exp(-1*r*T)*N2
        return P
```

```
In [6]: volatility = 0.28

        # Put Call Parity
        # C - P = S - K.exp(-rT)
```

```
P = current_value_put(S0, K, r, T, volatility)
print(f"Price of Put Option for given stock and volatility : ${P:.3f}")
```

Price of Put Option for given stock and volatility : \$0.932

A firm owns a patented drug that can be commercialized within the next 4 months. The launch involves a fixed regulatory and setup cost equivalent to

35million. The expected net revenue from the product launch (treated as the “stock”) is currently estimated at 38 million and follows a log-normal diffusion process. There is no intermediate revenue, and the firm can only make the launch decision now. Should the firm launch the product? Justify using a real options perspective and your results from (a) and (b). Assume the launch opportunity resembles a European call option.



```
In [7]: # The real options approach justifies the launch only if the option value of the launch exceeds
value = current_value_call(S0, K, r, T, volatility)

if (value >= S0 - K):
    print(f"Firm should launch as Current Value(${value:.2f} million) >= S-K(${3} million)")
else:
    print(f"Firm should not launch as Current Value(${value:.2f} million) <= S-K(${3} million)")
```

Firm should launch as Current Value(\$4.62 million) >= S-K(\$3 million)